

Fraud Detection System (Flask + SQLite)

A college-project demonstration of a rule-based transaction risk checker. It provides dashboard statistics, history stored in SQLite and Chart.js analytics.

Setup

1. Install Python 3.10 or newer.
2. Open the project folder in VS Code.
3. Create and activate a virtual environment (recommended):

```
python -m venv .venv
# Windows PowerShell
.venv\Scripts\Activate.ps1
# macOS/Linux
source .venv/bin/activate
```

4. Install Flask:

```
pip install -r requirements.txt
```

5. Start the application:

```
python app.py
```

6. Visit <http://127.0.0.1:5000> in a browser.

Rules

- Amount > ₹50,000: suspicious.
- More than 5 transactions/minute: fraud.
- Login country differs from transaction country: suspicious.
- Account less than 7 days old and amount > ₹20,000: fraud.

Important

This is an academic rule-based demonstration. Labels are not proof of fraud and must be reviewed by a qualified person before any account decision. The local SQLite database (fraud_detection.db) is created automatically on first run. Delete it only if you want to clear the test history.